

Software Testing

Selenium Automation Testing

Selenium Webdriver

Selenium WebDriver

<https://www.selenium.dev/documentation/webdriver/>



Selenium WebDriver



Selenium IDE



Selenium Grid

Download Apache NetBeans

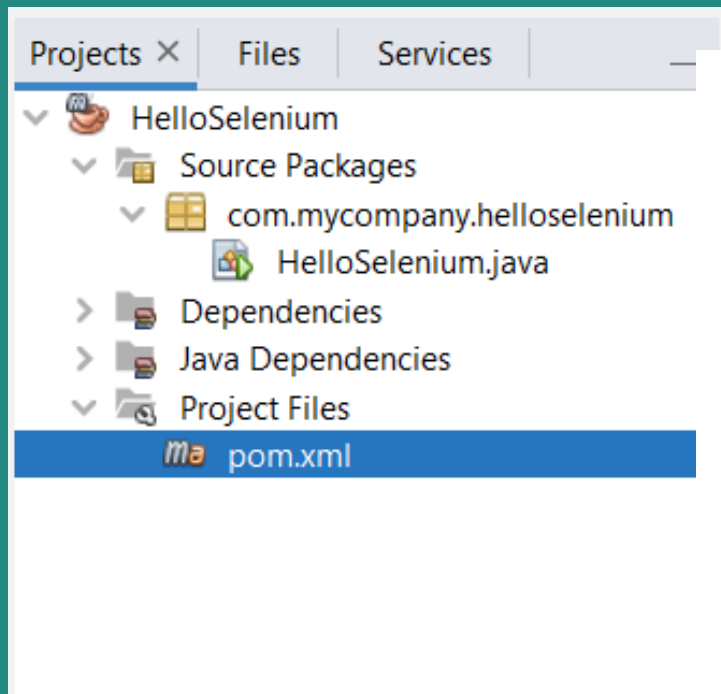
- <https://netbeans.apache.org/front/main/download/index.html>



Apache NetBeans

Install Selenium library

- https://www.selenium.dev/documentation/webdriver/getting_started/install_library/



Requirements by language

Java

Python

CSharp

Ruby

JavaScript

Kotlin

View the minimum supported Java version [here](#).

Installation of Selenium libraries for Java is accomplished using a build tool.

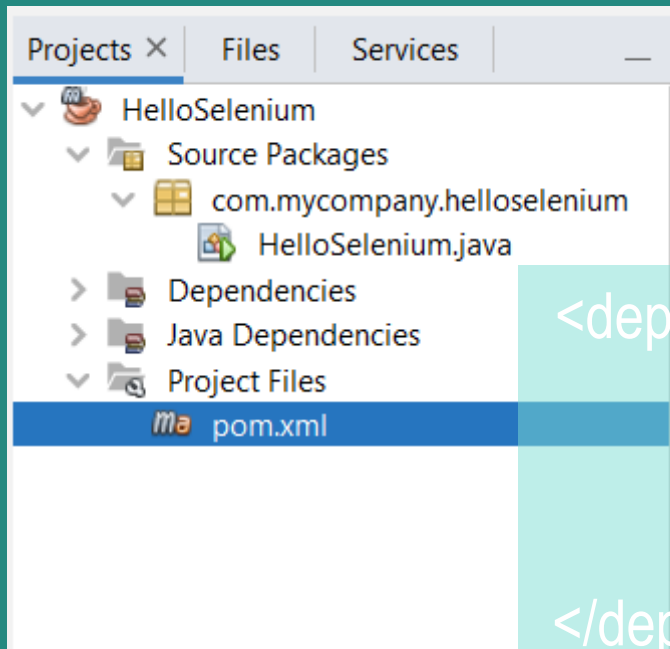
Maven

Specify the dependencies in the project's `pom.xml` file:

```
<dependency>
  <groupId>org.seleniumhq.selenium</groupId>
  <artifactId>selenium-java</artifactId>
  <version>${selenium.version}</version>
</dependency>
```

Install Selenium library

- Create Java *Maven* Project
- Specify the dependencies in the project's *pom.xml* file



```
<dependency>
```

```
<groupId>org.seleniumhq.selenium</groupId>
```

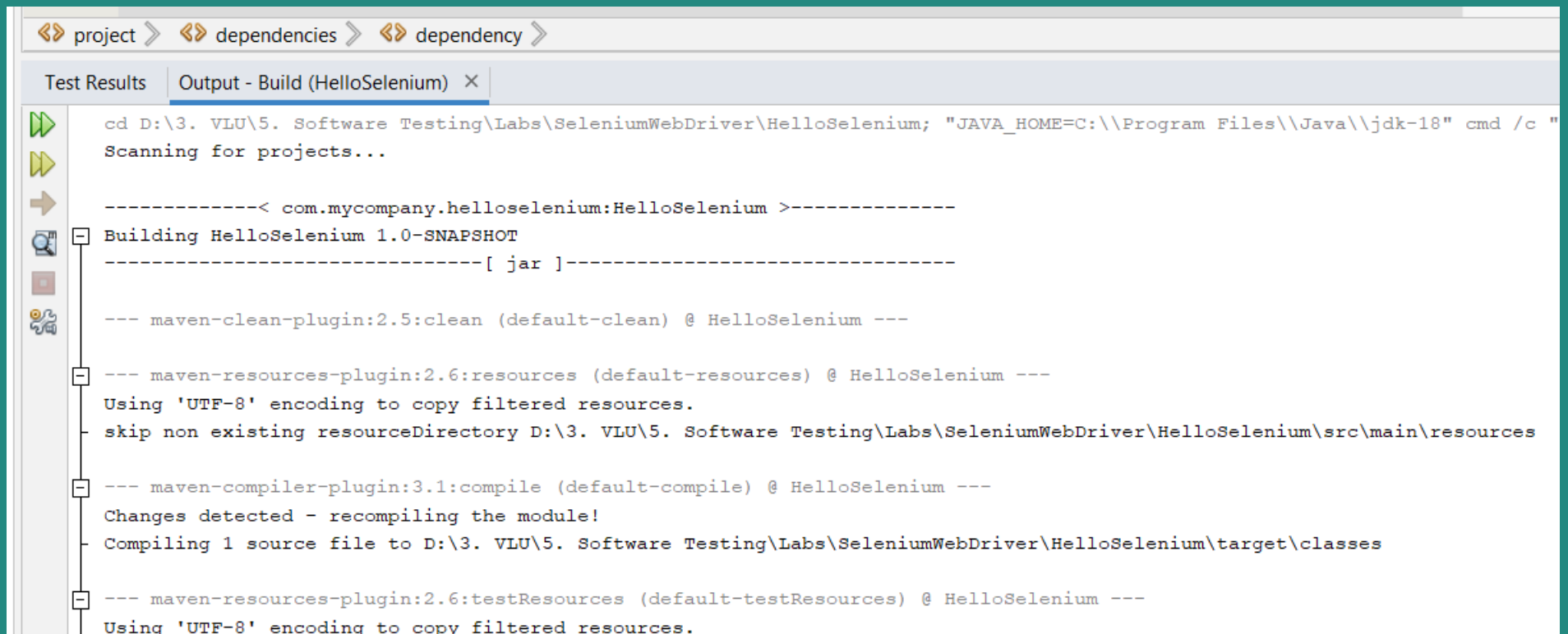
```
<artifactId>selenium-java</artifactId>
```

```
<version>4.18.1</version>
```

```
</dependency>
```

Build Project

■ maven-install-plugin



The screenshot shows an IDE window with a breadcrumb trail at the top: project > dependencies > dependency >. Below this is a tab bar with 'Test Results' and 'Output - Build (HelloSelenium) X'. The 'Output' tab is active, displaying the following build log:

```
cd D:\3. VLU\5. Software Testing\Labs\SeleniumWebDriver\HelloSelenium; "JAVA_HOME=C:\\Program Files\\Java\\jdk-18" cmd /c "
Scanning for projects...

-----< com.mycompany.helloselenium:HelloSelenium >-----
Building HelloSelenium 1.0-SNAPSHOT
-----[ jar ]-----

--- maven-clean-plugin:2.5:clean (default-clean) @ HelloSelenium ---

--- maven-resources-plugin:2.6:resources (default-resources) @ HelloSelenium ---
Using 'UTF-8' encoding to copy filtered resources.
skip non existing resourceDirectory D:\3. VLU\5. Software Testing\Labs\SeleniumWebDriver\HelloSelenium\src\main\resources

--- maven-compiler-plugin:3.1:compile (default-compile) @ HelloSelenium ---
Changes detected - recompiling the module!
Compiling 1 source file to D:\3. VLU\5. Software Testing\Labs\SeleniumWebDriver\HelloSelenium\target\classes

--- maven-resources-plugin:2.6:testResources (default-testResources) @ HelloSelenium ---
Using 'UTF-8' encoding to copy filtered resources.
```

Hello Selenium

```
5
6  package com.mycompany.helloselenium;
7
8  import org.openqa.selenium.WebDriver;
9  import org.openqa.selenium.chrome.ChromeDriver;
10
11 public class HelloSelenium {
12
13     public static void main(String[] args) throws InterruptedException {
14         WebDriver driver = new ChromeDriver();
15
16         driver.get("https://selenium.dev");
17         Thread.sleep(5000);
18
19         driver.quit();
20
21     }
22 }
```

First script

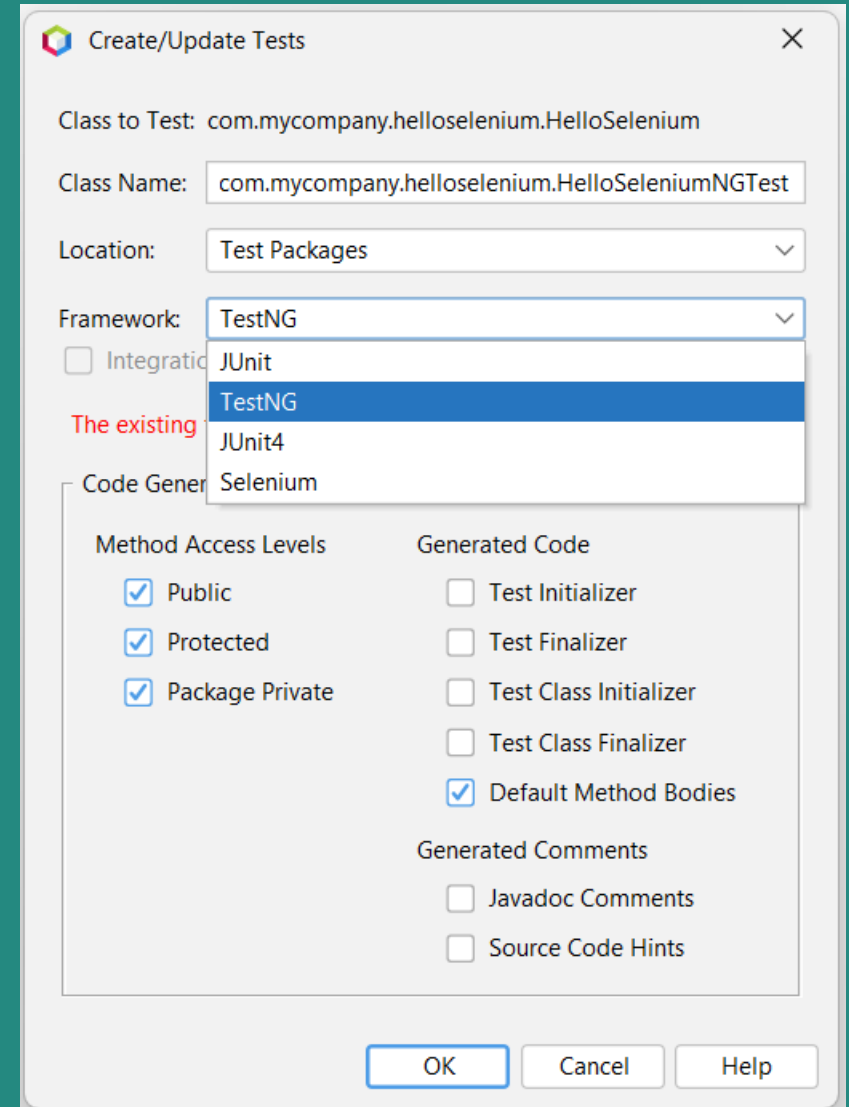
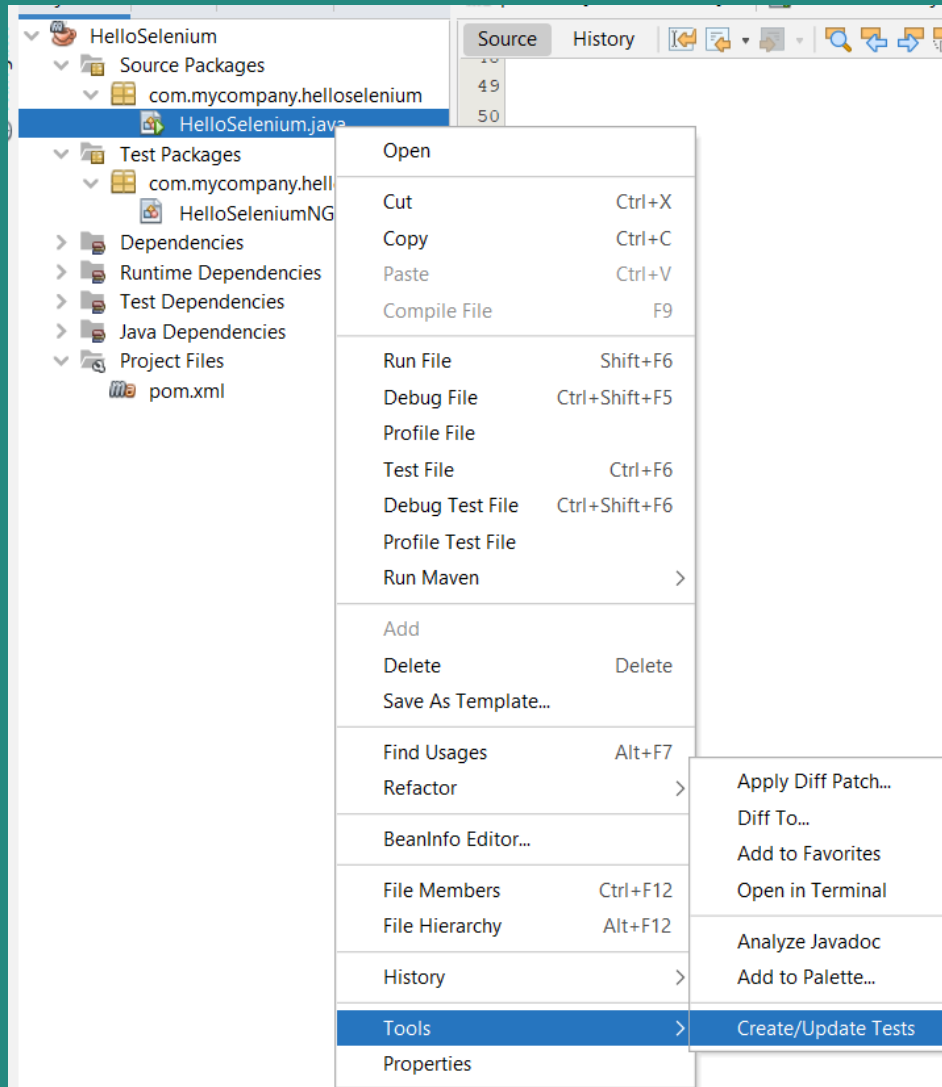
- https://www.selenium.dev/documentation/webdriver/getting_started/first_script/

```
19 WebDriver driver = new ChromeDriver();
20 String baseUrl = "http://www.demo.guru99.com/V4/";
21
22 // launch Firefox and direct it to the Base URL
23 driver.get(baseUrl);
24
25 // Enter username
26 driver.findElement(By.name("uid")).sendKeys("mngr1336");
27 Thread.sleep(4000);
28
29 // Enter Password
30 driver.findElement(By.name("password")).sendKeys("dAnavUq");
31 Thread.sleep(4000);
32
33 // Click Login
34 driver.findElement(By.name("btnLogin")).click();
35
36 Thread.sleep(4000);
37 driver.quit();
```


Second script

```
15 public static void main(String[] args) throws InterruptedException {
16     // Open Chrome browser
17     WebDriver driver = new ChromeDriver();
18     driver.get("https://www.selenium.dev/selenium/web/web-form.html");
19
20     // Get browser information:
21     driver.getTitle();
22     Thread.sleep(2000);
23
24     // Find an element
25     WebElement textBox = driver.findElement(By.name("my-text"));
26     WebElement submitButton = driver.findElement(By.cssSelector("button"));
27
28     // input data into an element
29     textBox.sendKeys("Selenium");
30     Thread.sleep(5000);
31     submitButton.click();
32
33     WebElement message = driver.findElement(By.id("message"));
34     message.getText();
35     Thread.sleep(2000);
36
37     driver.quit();
38 }
```

Create TestNG



Selenium TestNG

```
23      @Test
24      ▶  public void test01() throws InterruptedException {
25          WebDriver driver = new ChromeDriver();
26          driver.get("https://www.selenium.dev/selenium/web/web-form.html");
27
28          String title = driver.getTitle();
29          assertEquals("Web form", title);
30          Thread.sleep(2000);
31
32          WebElement textBox = driver.findElement(By.name("my-text"));
33          WebElement submitButton = driver.findElement(By.cssSelector("button"));
34
35          textBox.sendKeys("Selenium");
36          Thread.sleep(5000);
37          submitButton.click();
38
39          WebElement message = driver.findElement(By.id("message"));
40          String value = message.getText();
41          assertEquals("Received!", value);
42          Thread.sleep(2000);
43
44          driver.quit();
45      }
```

Locating GUI Elements

By.className	finds elements based on the value of the “class” attribute	<code>findElement(By.className(“someClassName”))</code>
By.cssSelector	finds elements based on the driver’s underlying CSS Selector engine	<code>findElement(By.cssSelector(“input#email”))</code>
By.id	locates elements by the value of their “id” attribute	<code>findElement(By.id(“someId”))</code>
By.linkText	finds a link element by the exact text it displays	<code>findElement(By.linkText(“REGISTRATION”))</code>
By.name	locates elements by the value of the “name” attribute	<code>findElement(By.name(“someName”))</code>
By.partialLinkText	locates elements that contain the given link text	<code>findElement (By.partialLinkText (“REG”))</code>
By.tagName	locates elements by their tag name	<code>findElement (By.tagName (“div”))</code>
By.xpath	locates elements via XPath	<code>findElement(By.xpath(“//html/body/div/table/tbody/tr/td[2]/table/tbody/tr[4]/td/table/tbody/tr/td[2]/table/tbody/tr[2]/td[3]/ form/table/tbody/tr[5]”))</code>

Commands

- `get(url)`
- `getTitle()`
- `getCurrentUrl()`
- `getText()`
- `navigate().forward()`
- `navigate().back()`
- `navigate().to()`
- `navigate().refresh()`
- `close()` close a single window;
- `quit()` close all windows
- `switchTo().frame()`
- `switchTo().alert()`
- **Implicit waits**
- **Explicit waits**

- `isEnabled()`
- `isDisplayed()`
- `isSelected()`

(if the element does not exists
WebDriver will throw
a *NoSuchElementException*)

```
try {
    WebElement ele = driver.findElement(By.id("username"));
    ele.sendKeys("ntthanh");
} catch (NoSuchElementException ex) {
    System.out.println("NoSuchElementException");
}
```

```
WebDriverWait myWaitVar = new WebDriverWait(driver, 3);
try {
    myWaitVar.until(ExpectedConditions.visibilityOfElementLocated(By
        .id("username")));
    driver.findElement(By.id("username")).sendKeys("tutorial");
} catch (TimeoutException toe) {
    System.out.println(toe.toString());
}
```

Alert

```
public static void main(String[] args) throws InterruptedException {  
    ChromeDriver chromedriver = new ChromeDriver();  
    chromedriver.get("https://output.jsbin.com/enifaf");  
    Thread.sleep(3000);  
  
    // Tìm button ở trang web  
    chromedriver.findElement(By.cssSelector("button")).click();  
  
    // Chuyển sang giao tiếp với alert  
    org.openqa.selenium.Alert alert = chromedriver.switchTo().alert();  
  
    // Kiểm tra dòng text trên alert  
    String alertText = alert.getText();  
    System.out.println(alertText);  
    Thread.sleep(3000);  
  
    // Nhấn OK trên alert  
    alert.accept();  
}
```

Interacting with web elements

- **click()**
- **sendKeys()**
 - types the keys into an element.
- **clear()**
 - resets the content of an element

Exercises

- Thực hành các thao tác click, send keys và clear
 - <https://www.selenium.dev/selenium/web/inputs.html>

TC1. Verify the title

```
// Open URL: https://demo.guru99.com/v4/

WebElement usernameInput = driver.findElement(By.id("username"));
WebElement passwordInput = driver.findElement(By.id("password"));

WebElement loginButton = driver.findElement(By.id("login-button"));

usernameInput.sendKeys("your_username");
passwordInput.sendKeys("your_password");
loginButton.click();

String expTitle = "???";
String actTitle = "getTitle()";

assertEquals(actTitle, expTitle);
```

- User: mngr563687
- Password: UvAnerE

TC2. Verify Login successfully

TC3. Verify the message of alert

```
String actUrl = driver.getCurrentUrl();  
String expUrl = "https://demo.guru99.com/v4/manager/Managerhomepage.php";
```

```
// input invalid user or password  
  
btnLogin.click();  
Alert alert = driver.switchTo().alert();  
String actAlertMsg = alert.getText();  
alert.accept();
```

<https://www.saucedemo.com/>

Test case ID	Description	Expected Result
TC_01	Kiểm tra title của trang web	
TC_02	Đăng nhập thành công với tài khoản hợp lệ	Chuyển hướng đến trang inventory.html Kiểm tra URL
TC_03	Đăng nhập thất bại với tài khoản không hợp lệ	Hiển thị lỗi: <i>Sorry, this user has been locked out.</i>
TC_04	Đăng nhập thất bại khi bỏ trống username hoặc password	Nếu bỏ trống username: <i>Username is required</i> Nếu bỏ trống password: <i>Password is required</i>
TC_05	Đăng xuất thành công	Quay lại trang https://www.saucedemo.com/

```
public class FirstTestNGFile {
    public String baseUrl = "http://pltpro.net";
    String driverPath = "C:\\\\chromedriver.exe";
    public WebDriver driver;

    @BeforeTest
    public void launchBrowser() throws InterruptedException {
        System.out.println("CHẠY TRÌNH DUYỆT CHROME");
        System.setProperty("webdriver.chrome.driver", driverPath);
        driver = new ChromeDriver();
        driver.get(baseUrl);
        driver.manage().window().maximize();
        Thread.sleep(2000);
    }

    @Test
    public void KiemThuSoSanhTieuDe() {
        String expectedTitle = "PLT SOLUTIONS (Panda Laptop 1)";
        String actualTitle = driver.getTitle();
        Assert.assertEquals(actualTitle, expectedTitle);
        if (actualTitle.toString().equalsIgnoreCase(expectedTitle.toString())) {
            System.out.println("HAI TIÊU ĐỀ MONG ĐỢI VÀ THỰC TẾ GIỐNG NHAU - PASS RỒI NHA");
        }
    }

    @AfterTest
    public void terminateBrowser() {
        driver.close();
    }
}
```